



Estd: 1986

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SJC INSTITUTE OF TECHNOLOGY

Chickballapur – 562 101

Department of Artificial Intelligence and Data Science

Tutorial - 2

SUBJECT TITLE	Big Data Analytics		
SUBJECT TYPE	Integrated		
SUBJECT CODE	BAD601		
ACADEMIC YEAR	2024-25 (Even semester)	Batch	
SCHEME	NEP CBCS 2022		
SEMESTER	6th sem, AI&DS		
FACULTY NAME and DESIGNATION	Darshan A, Assistant Professor		
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Q. No.	Questions	Level	COs
1	Explain the mapreduce daemons and also explain the working of mapreduce.	L2	CO2
2	Explain the concept of hadoop 1.x and hadoop 2.x with a neat diagram. Also explain daemons YARN and YARN architecture with a neat diagram.	L2	CO2
3	Explain the role of Mapper and Reducer tasks in detail. Write a mapreduce program to implement a matrix multiplication.	L3	CO2
4	Explain the role of combiner in mapreduce jobs and write an example program explaining the same.	L3	CO2
5	Develop a mapreduce program to implement wordcount by taking input from the text file.	L6	CO2
6	Develop a mapreduce program to find the tags associated with each movie by analyzing movie lens data.	L6	CO2
7	List out and explain the features of mongoDB. Write a queries to create a database named “db” and add a collection named “worker”, and add 10 documents into it where each document should have the fields name, age, place, salary. If the worker by the name "James" is not present in the database, then add the worker information and if the "James" exist, then update his salary to 30000. If the "Rohan" worker document is not existing, then add that document with the new filed as contactinfo to the mentioned files above. Update the 3rd worker age to 45 and display all the documents. Remove the place field from the 5th employee of the database.	L3	CO3
8	List out the differences between mongoDB and RDBMS. Write a queries for the below given problem statements: a) create a database b) drop a database c) list out the collections	L3	CO3

	d) create collection e) insert documents into the collection students and each document should have fields such as rollno, age, contactno, email id. f) Display the documents from the collection		
9	Write a query for the below given statements: a) To search for documents from the “students” collection based on search criteria, name should be “royal” b) To display the studentname from all the documents of the students collection. The identifier “_id” should be suppressed and NOT displayed. c) To display the studName, grade as well the identifier, _id from the document of the students collection where the _id column is 1. d) To find those documents where the grade is NOT set to ‘VII’. e) To find those documents from the students collection where the hobbies is set to either ‘chess’ or is set to ‘skating’. f) To find the documents from the students collection where the studname begins with ‘M’. g) To find documents from students collection where the StudName ends in ‘s’. h) Add null values to a field location whose ids are _id:3 and _id:4 i) Display the documents which has location value as null. j) To remove location field having “Null” values from the documents (_id:3 and _id:4) from the students collection.	L3	CO3
10	Explain and write an example queries for each of the given below: a) Count b) Sort c) Limit d) Skip e) Aggregate Using mongoDB	L3	CO3
11	Develop a mapreduce program that mines weather data and displays appropriate messages indicating the weather conditions of the day.	L6	CO2
12	Explain the sharding and replication feature of mongoDB with a neat block diagram.	L2	CO3
13	Develop a program to illustrate the concept of cursors in MongoDB.	L6	CO3
14	Develop a program to illustrate the concept of MongoImport and MongoExport with an example program.	L6	CO3

Sign of the faculty
Darshan A

HOD